

If You Build It, They Will Learn: The Effect of Construction Spending on Construction Trades Graduates

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Abstract

We examine the impact of construction activity on construction trade degrees and find lagged declines in trade degree completions associated with increased activity. Using county-level construction spending data matched with institution-level degree conferral data from the Integrated Postsecondary Education Data System (IPEDS) for non-metropolitan U.S. counties from 2010 to 2019, we estimate fixed-effects regression models that account for persistent differences across institutions, counties, and years, and allows for lagged effects. We split the sample by spending quartiles. We find a 1% increase in spending leads to an immediate 6% decline in degree completions and a 10% decline two years later for the bottom quartile. The top quartile sees 16% decline one year after a 1% increase in spending somewhat mitigated by a 10% increase two years after.

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1 Introduction

Despite the economic uncertainty associated with tariff impacts (Associated General Contractors of America, 2025), and the sensitivity of construction companies to that uncertainty, the construction industry remains a strong source of employment opportunities. Demand for skilled workers and construction professionals at every level remains consistently high. The Associated General Contractors of America (AGC) published the results of its annual workforce shortage survey and found that in 2024, 94% of construction firms were having difficulty filling positions (Associated General Contractors of America, 2024). In the United States, the construction sector added 11,000 jobs in April 2025 as firms geared up for the busy season (Associated General Contractors of America, 2025). The U.S. Bureau of Labor Statistics reported 248,000 construction job openings and 348,000 new hires in April 2025 (U.S. Bureau of Labor Statistics, 2025). Although job openings declined by 24% from an unusually active 2024, the number of open positions continues to suggest favorable employment opportunities at living wages.

As demand for skilled workers continues in the construction industry, enrollment in skilled-trades programs has also increased (Steinberg and Nadworny, 2022; Sanchez, 2023). Nationally, construction trades enrollment increased by 24.1% from fall 2022 to fall 2024 (National Student Clearinghouse Research Center, 2024b). However, a concerning trend has emerged regarding retention, or degree and certificate completion rates, within construction trade programs. The 2024 report from the National Student Clearinghouse Research Center reported a four-year average retention rate of only 59.75% for associate degrees and 56.45% for undergraduate certificates (National Student Clearinghouse Research Center, 2024a).

Traditionally, on-the-job training has been sufficient for early-career construction workers, providing a foundation that has supported entrepreneurship in the industry for over a century. However, formal training creates opportunities for workers to acquire advanced skills and achieve accelerated career placement (Killingsworth et al., 2012). In today's construction industry, tradecraft has become increasingly complex as methods, materials, and technologies continue to advance. The application of engineered systems requires deeper understanding of complex mechanical and electrical building systems, Building Information Modeling (BIM), LiDAR, 3D printing, and modular construction systems (Killingsworth et al., 2021). Consequently, as the intricacy of construction work increases, the demand for formal certificates and academic degrees should also increase (Mohammadi et al., 2019).

Considering the demand for construction workers across the country it seems rea-

sonable to assume an equally high demand for trade, craft, and construction skills training. Thus, a guiding question for this study was: do we observe a consistent relationship between construction spending and educational attrition?

This study first examined the relationship between construction spending, as a proxy for labor demand, and certificate and degree completion rates in trade programs, as a proxy for labor supply. The findings — which are generated from observable trends in two datasets — are informative to individuals considering a construction career, to academics training those individuals, and to the industry in which those individuals are ultimately employed.

2 Literature Review

2.1 Associative Research Literature

In November 2021, the Biden–Harris administration enacted the Bipartisan Infrastructure Law, which infused nearly \$700 billion into state and local infrastructure projects throughout the country. According to the U.S. Department of Transportation, approximately 670,000 construction jobs were created as a result of this legislation (U.S. Department of Transportation, 2023). Despite the increased demand for labor, the supply of degree completers in construction trade programs has not responded proportionately. In 2018, AGC reported that 80% of construction firms experienced difficulty filling hourly craft positions (Associated General Contractors of America, 2018). Following the passage of the infrastructure bill, the percentage of firms reporting workforce vacancies increased to 94% (Associated General Contractors of America, 2024).

Research on attrition in trade-training programs has a long history. Wagner and Kulwiec (2022) examined attrition in union-sponsored pre-apprenticeship programs, with a specific focus on women in construction. Their qualitative findings indicated that partnerships between training programs and employers encouraged program completion, while misalignment between training content and job-site expectations contributed to attrition.

Earlier work by Bilginsoy (2003) investigated attrition and retention in construction apprenticeship programs and found that joint academic–employer-sponsored programs were more effective at reducing attrition. A key conclusion of that study was that skilled labor shortages were already a significant concern, yet empirical research examining the relationship between academic retention and apprenticeship structures

remained limited. This gap remains relevant today and is directly addressed in the present study.

Attrition persisted in global trade-training programs through the mid-2000s. Snell and Hart (2007) identified concerns related to training quality compared to on-the-job learning. Bilginsoy (2017) later analyzed attrition trends over a thirty-year period and found that patterns remained largely unchanged since the 1990s. Consistent with earlier findings, joint employer–trainer partnerships were associated with higher retention and completion rates.

More recently, Zhang et al. (2025) examined the relationship between macroeconomic indicators and construction spending, identifying delayed effects between predictive variables and spending outcomes. Their findings provide a useful economic framework for examining lagged relationships between demand-side construction indicators and supply-side workforce outcomes, which directly informs the analytical approach used in this study.

3 Methodology

3.1 Data

The data for this paper cover 2010-2019 and are made up of two parts: proprietary data on aggregate construction spending by county-year for all non-metro counties in the United States (provided by Industrial Info Resources) and annual construction trades degree completion data from the Integrated Postsecondary Education Data System (IPEDS) Completions Surveys.

Industrial Info Resources (IIR) collects data on several industries including: pipelines, food & beverage, electric power, and more. IIR has been operational since 1983 and bills itself as “Your Trusted Data Source for Industrial Market Insight”. The data was acquired using a grant from *The Moore School Research Grant Program*.

The IPEDS Completions Survey is a key federal dataset administered annually by the National Center for Education Statistics (NCES). IPEDS surveys all postsecondary institutions participating in federal student aid programs and therefore provides near-universal coverage of US technical colleges, community colleges, colleges, and universities. The survey collects detailed information on the number and types of degrees and certificates conferred by accredited postsecondary institutions. These data are disaggregated by award level, gender, race/ethnicity, and most importantly field of study as indicated by Classification of Instructional Programs (CIP) code.

Each CIP code corresponds to a specific field of study and includes a six-digit numerical identifier structured in a hierarchical format: the first two digits represent the broad discipline, the next two identify an intermediate group, and the final two specify a detailed program area. The CIP code 46.0102 is for bricklaying/bricklayer. The listed description is *a program that prepares individuals to lay brick and block and construct or repair walls, partitions, fireplaces, chimneys, and other structures*. The two digit series, 46, indicates construction trades, the two digit subseries, 01, indicates masonry, and the six digits refers specifically to bricklaying. Similarly, CIP code 46.0101 refers to general masonry—*a program that prepares individuals to lay brick, block, and other types of masonry units, and to construct, install, finish, and repair masonry structures. Includes instruction in brick and block laying; mortar and bonding agents; blueprint reading; layout; estimation; residential and commercial masonry techniques; construction technology; equipment operation and maintenance; safety; and applicable codes and standards*. The data are also disaggregated by award level. From this we are able to separate associate’s degree conferrals from bachelor’s degree conferrals.

Table 1 below highlights construction spending statistics for counties that were matched to the IPEDS data, i.e., those counties with at least one Construction Trades degree program during the sample period.

Table 1: Construction Spending by Year

Year	Mean	SD	Min	Max	N
2010	350.93	673.46	0.03	3 862.82	338
2011	361.12	788.70	0.51	4 687.09	331
2012	435.68	988.54	0.56	5 923.72	330
2013	366.17	766.58	0.58	6 008.62	356
2014	372.77	826.62	0.00	8 281.63	363
2015	402.16	914.72	0.00	8 542.55	369
2016	402.47	835.90	0.00	7 118.10	381
2017	419.88	845.93	0.00	7 212.25	385
2018	454.91	878.17	0.00	7 282.67	383
2019	425.01	798.28	0.00	6 979.14	387

Notes: Spending values are reported in millions of dollars. N is the number of counties in that year with both proprietary construction spending data and at least one Construction Trades degree program.

Although annual averages hover between \$300 and \$500 million dollars over time, there is significant variation between counties. Aggregate construction spending ranges from less than \$1 million to several billion dollars in each year of the data.

Table 2 below provides summary statistics for the IPEDS degree conferral data. We identified construction trades programs as classified under the two digit CIP code 46 (Construction Trades). We focus on Associate’s degree programs, which are the large majority of construction trades programs. As in the spending data, relatively stable means over time mask significant variation between counties. Each year in our data includes counties with zero construction trades graduates, as well as counties with well over one hundred graduates.

Table 2: Construction Graduates by Year

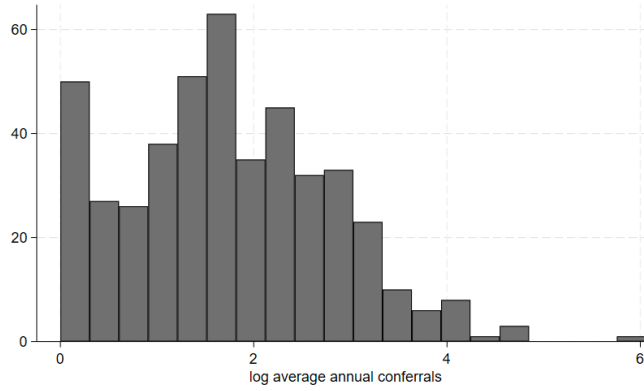
Year	Mean	SD	Min	Max	N
2010	10.51	17.67	0	170	338
2011	11.61	18.31	0	151	331
2012	14.12	40.10	0	671	330
2013	11.52	31.35	0	531	356
2014	11.00	26.72	0	408	363
2015	10.27	24.01	0	374	369
2016	10.18	26.43	0	424	381
2017	11.74	32.37	0	503	385
2018	11.92	28.61	0	463	383
2019	12.56	34.67	0	580	387

Notes: We define ‘Construction Graduates’ as recipients of an Associate’s degree from a program classified under CIP code 46 (Construction Trades) between 2010 and 2019. Yearly means are at the county level. N is the number of non-metro counties with graduation data in that year.

As can be seen in Table 2, the amount of conferrals varies widely across counties, also visible in Figure 1.

Figure 1 shows the distribution of average annual conferrals across institutions using the natural log, which addresses a predictable feature of the raw data. The concentration of observations at lower levels of degree conferral reflects the structure of construction trades education in the United States. Many institutions in our sample are small community and technical colleges serving local labor markets and conferring relatively few construction trades degrees each year, though a small number of large vocational institutions account for the upper tail. For example, Trinidad State

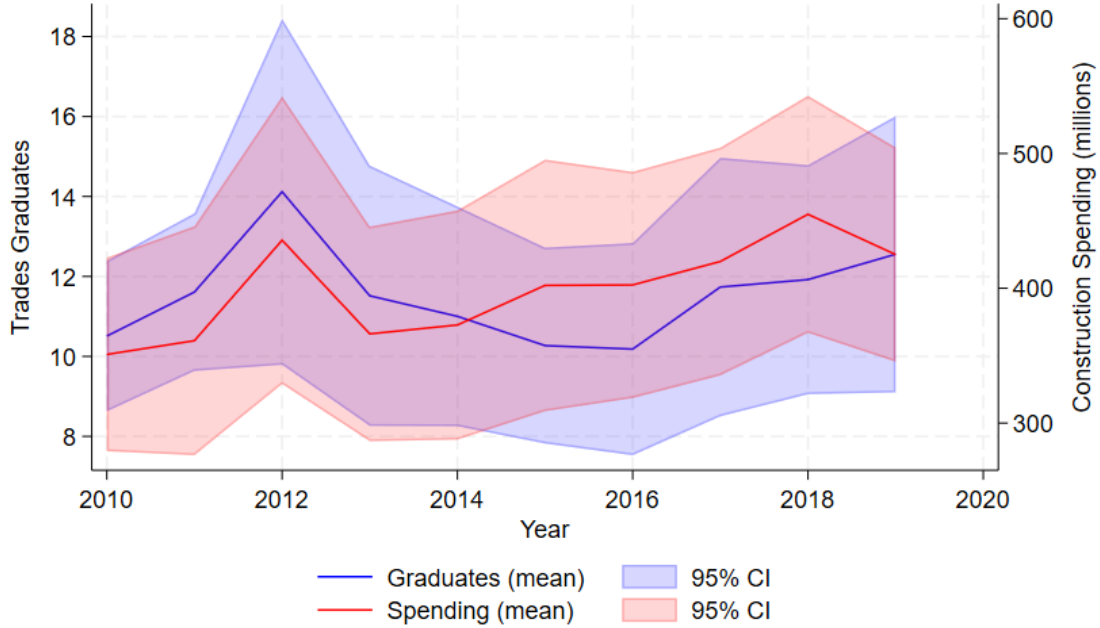
Figure 1: Mean Yearly Graduates by School (logged)



Note: The x-axis is the logged mean number of graduates degrees per year by school. The y-axis shows the number of schools with that value.

Junior College in Colorado typically confers between 0-2 degrees each year, while Red Rocks Community College conferred at least 23 degrees annually during our sample. This heterogeneity motivates the log transformation analytically: simply using the raw numbers would not account for the fact that a one-unit increase in conferrals would represent a much larger proportional shift for schools like Trinidad State (a 100% increase if prior year had 1 conferral) than for schools like Red Rocks (just a 4.3% increase if prior year had 23 conferrals). Using the natural log converts these level changes into percentage changes, which puts institutions of different sizes on more equal footing. Consequently, a low mean value of construction graduates per year with a large standard deviation (see Table 2) is expected. All regression analyses likewise use the natural log to account for this scale heterogeneity across institutions.

Figure 2: Construction Spending and Trades Graduates



Note: The y-axis on the left represents average annual construction trades degree completions, while the right y-axis represents average annual construction spending in millions (both at the county level). Trends in graduates and spending over time are shown in blue and red, respectively.

Figure 2 shows the means and 95% confidence intervals for our matched sample over time.

3.2 Two-Way Fixed Effects Panel Models

Our data contains observations of nearly 400 U.S. counties from 2010–2019, inclusive, and can be viewed as a combination of cross-sectional and time-series information, commonly referred to as panel data. We expect systematic differences across counties arising from characteristics that are not directly observed in the data. For example, counties with a long-standing presence of construction firms or established supply chains may consistently exhibit higher levels of construction spending. In this case, a standard ordinary least squares analysis (Greene, 2012, Ch. 4; Wooldridge, 2002, Ch. 2) would yield results that are contaminated by omitted variable bias Greene,

2012, Ch. 11; Wooldridge, 2002, Ch. 10), as persistent county characteristics—such as long-standing industry presence—would be incorrectly attributed to changes in construction spending. If instead we were to group the data by county and subtract each county’s mean value from its observations, a process called de-meaning, counties would be compared to their own historical averages rather than to other counties with fundamentally different baseline conditions. This can be done by including a county fixed effect, similar to the individual fixed effects approach used by Onayev et al. (2022) in their analysis of highway construction prices across U.S. states. In their setting, fixed effects isolate how changes in cost drivers affect infrastructure prices while controlling for persistent state characteristics, which parallels our goal of examining how changes in construction spending relate to educational outcomes while controlling for persistent county characteristics.

In economic studies, it is important to expect systematic differences over the years. For example, recovery from the Great Recession led to sustained enrollment declines from 2013-2019 while the 2015 Fixing America’s Surface Transportation (FAST) Act was a nationwide investment in the construction sector. Both influenced all counties but not all years. Failing to account for these time-specific national shocks (i.e. something that disrupts the normal pattern of economic activity) could cause nationwide trends to be mistakenly attributed to county-level changes in construction activity and enrollment. This can be addressed by grouping the data by year and subtracting the yearly mean from each observation—de-meaning across years—which compares variation within a given year to that year’s overall average. Mechanically, this is handled by including time fixed effects in our model.

The model used in this study included both county and year fixed effects, a specification commonly referred to as a two-way fixed effects (TWFE) model Greene, 2012, pg. 363; Wooldridge, 2002, pg. 254. Such models are workhorses for analyzing panel data because they simultaneously control for unobserved differences across units and common shocks over time. Xu et al. (2009) provides an introduction to panel models, highlighting the usefulness of the TWFE framework for analyzing highway accidents in China when unobserved, time-invariant characteristics vary across locations. More recent studies apply similar TWFE approaches in closely related contexts. Pan et al. (2025) use county and year fixed effects to examine how green space quality affects fiscal stress, explicitly accounting for persistent regional differences and nationwide temporal trends. Cheng and Hu (2026) employ a TWFE model in the *Journal of Construction Engineering and Management* to study the impact of digital transformation on corporate performance, relying on within-firm variation over time while controlling for common macroeconomic shocks.

TWFE models can be implemented with ease using modern software. *Stata* users have traditionally used *xtreg* (<https://www.stata.com/manuals/xtxtreg.pdf>) though *reghdfe* (https://www.stata.com/meeting/chicago16/slides/chicago16_correia.pdf) is more common now. For *R* users, the *lfe* package provides the *felm* command (<https://www.rdocumentation.org/packages/lfe/versions/3.1.1/topics/felm>), however the *feols* command from the *fixest* package (Bergé, 2018) (<https://cran.r-project.org/web/packages/fixest/index.html>) is faster and more versatile.

3.3 Analytical Approach

Our primary empirical specification examines how annual construction trades degree completions are associated with changes in construction spending within the same county over time, while accounting for persistent differences across institutions, counties, and years. To do this, we estimate an ordinary least squares regression that includes institutions, county, and year fixed effects. We also include lagged measures of construction spending because degree completion reflects decisions made over multiple years, and labor market conditions experienced earlier in a student’s enrollment may influence whether they persist to completion or exit for employment. The general functional form for this specification is as follows:

$$\ln[Grads_{ict}] = \alpha + \theta_i + \gamma_c + \delta_t + \epsilon_{ct} + \sum_{j=0}^{p \in \{0,1,2\}} \beta_j \ln[Spending_{c,t-j}] \quad (1)$$

where $\ln[Grads_{ict}]$ is the natural log of the number of Associate graduates with degrees in construction trades at institution i in county c in year t .

α is not of substantive interest because it represents the predicted number of degree completions when construction spending is zero and all fixed effects are set to their reference values—a scenario that does not correspond to a meaningful or observable real-world condition. ϵ_{ct} is the idiosyncratic error term.

θ_i , γ_c , and δ_t are institution, county, and year fixed effects, respectively. Technically, fixed effects operate by de-meaning the data along each fixed-effect dimension. Institution fixed effects subtract each institution’s average level of degree conferrals across all years, county fixed effects subtract each county’s average level across all years, and year fixed effects subtract common national averages in each year. After these transformations, the remaining variation reflects deviations from an institution’s typical outcomes, net of persistent county characteristics and common time

trends. As a result, the analysis relies on changes over time within the same institution and county rather than comparisons across different institutions, counties, or years. This approach helps mitigate bias from omitted variables that are constant within institutions or counties over time—such as program quality, institutional mission, or long-standing local economic conditions—by removing their influence from the estimated relationships.

Construction spending could induce people to initiate a degree. As we are focused on degree completions, which could take some years, we include lagged effects of construction spending—up to two years. This means spending from, for example, 2015 can affect degrees granted in 2017. This is indicated in Equation (1) by the summation over previous years, $\sum_{j=0}^{p \in \{0,1,2\}} \beta_j \ln[Spending_{c,t-j}]$. Allowing p to take on the values of 0, 1, and 2 generates Equation (2) through Equation (4), respectively.

$$\ln[Grads_{ict}] = \alpha + \theta_i + \gamma_c + \delta_t + \varepsilon_{ict} + \beta_0 \ln(Spending_{ct}) \quad (2)$$

$$\begin{aligned} \ln[Grads_{ict}] = \alpha + \theta_i + \gamma_c + \delta_t + \varepsilon_{ict} &+ \beta_0 \ln[Spending_{c,t}] \\ &+ \beta_1 \ln[Spending_{c,t-1}] \end{aligned} \quad (3)$$

$$\begin{aligned} \ln[Grads_{ict}] = \alpha + \theta_i + \gamma_c + \delta_t + \varepsilon_{ict} &+ \beta_0 \ln[Spending_{c,t}] \\ &+ \beta_1 \ln[Spending_{c,t-1}] \\ &+ \beta_2 \ln[Spending_{c,t-2}] \end{aligned} \quad (4)$$

We take the natural logarithm of graduates and construction spending to the scale-dependent variation. This transforms level changes, like a spending change of \$10 million which may be significant to a small county but not to a large one, into percentage changes. With this, we are able to interpret the coefficients as:

- β_0 as the percentage change in the number of trade associate degrees resulting from a 1% increase in construction spending in the same year
- β_1 as the percentage change in the number of trade associate degrees resulting from a 1% increase in construction spending in the previous year
- β_2 as the percentage change in the number of trade associate degrees resulting from a 1% increase in construction spending two years ago

4 Results

Table 3 shows the effects of construction spending on the number of graduates of construction trades programs.

Table 3: The Effect of Construction Spending on Associate Degree Completion

	(1)	(2)	(3)
Logged spending	-0.030 (0.024)	-0.027 (0.026)	-0.043 (0.028)
1-year lag		-0.044 (0.028)	-0.035 (0.031)
2-year lag			-0.021 (0.036)
Observations	3041	2593	2203
Schools	417	385	357

Notes: Standard errors in parentheses. All three models represent regressions of logged construction spending on logged construction trades graduates. All models include county, year, and institution fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3 shows no statistically significant impact. However, the point estimates are all negative, suggesting that a 1% increase in construction spending *decreases* the number of graduates with associate's degrees in construction trades. Our preferred specification has two years of spending lags and is shown in column (3). We find that a 1% increase in construction spending in a given year is associated with 0.043% fewer construction trades graduates, though this is not statistically significant at any conventional level. The estimated effects of lagged spending are also negative and not statistically significant for both years. Table 4 splits the sample into the top and bottom quartiles of construction spending based on county spending in 2010. Column (3) indicates that a 1% increase in construction spending is associated with a decrease in construction trades graduates among institutions located in counties in the first quartile of 2010 spending by 0.06%, which is significant at the 10% level. The coefficient on 2-year lagged spending indicates 0.109% fewer graduates in response to

a 1% spending increase two years prior. Column (6) shows results for counties in the top quartile of spending, where we find a 1% increase in 1-year lagged spending is associated with a statistically significant (5% level) 0.166% fewer construction trades graduates. For 2-year lagged spending, a 1% increase in spending is associated with .1% additional graduates, which is significant at the 10% level.

Table 4: Construction Spending on Trades Graduates - Spending Quartiles

	Bottom Quartile			Top Quartile		
	(1)	(2)	(3)	(4)	(5)	(6)
Logged spending	0.001 (0.039)	-0.039 (0.035)	-0.060* (0.035)	-0.072 (0.053)	-0.074 (0.067)	-0.090 (0.081)
1-year lag		-0.002 (0.044)	0.009 (0.040)		-0.046 (0.057)	-0.166** (0.064)
2-year lag			-0.109* (0.058)			0.100* (0.054)
Observations	533	458	397	982	839	720
Schools	70	65	64	126	116	108

Notes: Standard errors in parentheses. All six models represent regressions of logged construction spending on logged construction trades graduates. Columns (1)-(3) represent estimates for the counties that were in the bottom quartile of construction spending in 2010, while columns (4)-(6) are for counties that were in the top quartile. All models include county, year, and institution fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5 Conclusion

We examine the relationship between construction activity and construction trades associate degree completions using county-level panel data from 2010–2019. While aggregate models show no statistically significant relationship between construction spending and degree completions, the direction of the estimates is consistently negative, suggesting industry activity has a negative effect on completed degrees.

We disaggregated the sample into top and bottom (by quartile) counties for construction spending. In the bottom-spending county, we find a 1% increase in construction spending causes an immediate 6% decline in completed degrees and a 10% decline two years later. In higher-spending counties, increased spending is associated with

short-run declines of 16% in completions followed by a 10% completion rebound two years later. On average, schools in bottom-spending counties lose 1.5 completed degrees and schools in top-spending counties lose 2.6 completed degrees but incur a latent 1.6 bump in completed degrees.

These findings suggest that strong labor demand may pull students into employment before completing their degrees, particularly in areas where construction markets are expanding rapidly. In lower-spending counties, where educational and industry infrastructures may be thinner, increased job opportunities appear more likely to disrupt program completion. In counties with higher-spending, the temporary decline followed by partial recovery may reflect more established industry-to-education links that allow some students to return and complete their degrees.

From a workforce development perspective, the results highlight a tension between immediate labor demand and long-term human capital formation. While rising construction spending creates employment opportunities, it may simultaneously reduce formal credential attainment if students exit early for work. For educators and policymakers, this underscores the importance of strengthening partnerships between training programs and employers, designing flexible completion pathways, and developing policies that encourage degree completion even during periods of strong labor demand.

Overall, the findings contribute to a deeper understanding of how economic conditions in the construction sector interact with educational outcomes. As infrastructure investment and construction activity continue to fluctuate with macroeconomic and policy conditions, aligning workforce development strategies with industry cycles will be critical to sustaining both short-term labor supply and long-term industry productivity.

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